

Reviewer Pack — Minimal Symplectic Form Rank Bounds Resolution Proof Width

Entry 687e17de88a9 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Symplectic Form Rank Bounds Resolution Proof Width

Entry ID: 687e17de88a9 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-30 01:00:07 UTC

1. Conjecture

Field A (mathematical branch): Symplectic Geometry **Field B** (complexity object): Boolean Function Complexity: Resolution Proof Complexity

Statement:

For any unsatisfiable CNF φ with n variables, the minimal rank of its symplectic form, denoted as $r(\varphi)$, is upper-bounded by a constant c times the width of its tree-like resolution proof, $w(\varphi)$. Equivalently, for all instances φ , $|r(\varphi) - c \cdot w(\varphi)| \leq k$.

Rationale (proposer's reasoning):

Symplectic geometry provides a geometric framework to study the structure of vector spaces, which could reveal new insights into the combinatorial complexity of Boolean functions. The symplectic form invariant might capture hidden complexities in resolution proofs, offering a new angle for lower-bound studies.

Taxonomy category: cg_kw_andreev (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 54a7f9254e8f6d81

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal rank of the symplectic form $r(\varphi)$ for unsatisfiable CNF φ with n variables satisfies $|r(\varphi) - c \cdot w(\varphi)| \leq k$, where c is a constant and k is a specified threshold.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	1.00	UNCERTAIN	SAFE

4. Novelty audit

Verdict: ADJACENT_OK against 6 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - symplectic geometry AND resolution proof complexity - minimal symplectic form rank bounds CNF resolution width - Boolean function complexity upper bound on symplectic geometry form rank

Top relevant hits considered: - [<http://arxiv.org/abs/1612.04764v1>] Cohomological aspects on complex and symplectic manifolds - [<http://arxiv.org/abs/1811.09984v4>] Translated points for contactomorphisms of prequantization spaces over monotone symplectic toric manifolds - [<http://arxiv.org/abs/1002.3099v3>] Tamed Symplectic forms and SKT metrics - [<http://arxiv.org/abs/math/0601540v1>] Symplectic forms and surfaces of negative square - [<http://arxiv.org/abs/1210.1135v2>] The closure of the symplectic cone of elliptic surfaces - [<http://arxiv.org/abs/1009.2975v3>] 2-plectic geometry, Courant algebroids, and categorified prequantization

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.4s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_random_cnf(n: int, m: int):
        cnf = []
        for _ in range(m):
            clause = [random.randint(1, n), -random.randint(1, n)]
            if len(set(clause)) == 2:
                cnf.append(clause)
        return cnf

    def tree_like_resolution_width(cnf):
        clauses = set(tuple(sorted(clause)) for clause in cnf)
        unit_clauses = {clause for clause in clauses if len(clause) == 1}

        while unit_clauses:
            lit, _ = unit_clauses.pop()
            new_clauses = []
            for clause in clauses:
                if lit in clause:
                    continue
                if -lit in clause:
```

```

        new_clause = tuple(sorted(set(clause) - {lit}))
        if len(new_clause) == 1:
            unit_clauses.add((new_clause[0],))
        else:
            new_clauses.append(new_clause)
    clauses.update(new_clauses)

    return len(clauses)

def symplectic_form_rank(cnf):
    n = max(abs(lit) for clause in cnf for lit in clause)
    matrix = [[0] * (n + 1) for _ in range(n + 1)]

    for clause in cnf:
        for i, lit in enumerate(clause):
            if i == len(clause) - 1:
                continue
            j, lit2 = clause[i + 1], -lit
            matrix[abs(lit)][abs(lit2)] += 1
            matrix[abs(lit2)][abs(lit)] += 1

    rank = 0
    for i in range(1, n + 1):
        pivot = next((j for j in range(i, n + 1) if matrix[j][i] != 0), None)
        if pivot is None:
            continue
        rank += 1
        for j in range(n + 1):
            matrix[i][j], matrix[pivot][j] = matrix[pivot][j], matrix[i][j]
        for j in range(1, n + 1):
            if i != j:
                factor = Fraction(matrix[j][i], matrix[i][i])
                for k in range(n + 1):
                    matrix[j][k] -= factor * matrix[i][k]

    return rank

def spearman_rank_correlation(x, y):
    n = len(x)
    x_ranks = {x[i]: i for i in range(n)}
    y_ranks = {y[i]: i for i in range(n)}
    x_sorted = sorted(x_ranks.keys())
    y_sorted = sorted(y_ranks.keys())

    rank_x = [x_ranks[val] for val in x_sorted]
    rank_y = [y_ranks[val] for val in y_sorted]

    n = len(rank_x)
    sum_d1 = sum((rank_x[i] - rank_y[i]) ** 2 for i in range(n))
    sum_d2 = sum((rank_x[i] - (i + 1)) ** 2 for i in range(n)) + sum((rank_y[i] - (i + 1)) ** 2 for i in range(n))

    rho = 1 - 6 * sum_d1 / sum_d2
    return rho

n = random.randint(5, 40)
m = random.randint(2 * n, 3 * n)

```

```

cnf = generate_random_cnf(n, m)

w_phi = tree_like_resolution_width(cnf)
r_phi = symplectic_form_rank(cnf)

if w_phi == 0:
    return {
        "metric_name": "Spearman rank correlation",
        "metric_value": None,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": "w( $\phi$ ) = 0, cannot compute -r( $\phi$ )"
    }

rho = spearman_rank_correlation([-r_phi], [math.log2(w_phi)])

return {
    "metric_name": "Spearman rank correlation",
    "metric_value": rho,
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": abs(rho) > 0.5,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    if all("metric_value" in r and r["metric_value"] is not None for r in results):
        mean_rho = sum(r["metric_value"] for r in results) / len(results)
        std_rho = math.sqrt(sum((r["metric_value"] - mean_rho) ** 2 for r in results) / len(results))
        support_fraction = sum(1 for r in results if abs(r["metric_value"]) > 0.5) / len(results)

        if support_fraction >= 0.8:
            print(f"RESULT: SUPPORTED mean={mean_rho} std={std_rho} support_fraction={support_fraction}")
        else:
            print(f"RESULT: FALSIFIED counterexample=\"rho_threshold_not_met\" first_failing_seed={seeds[0]}")
    else:
        print("RESULT: INCONCLUSIVE some_rho_values_none")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

'n_max': 27, 'conjecture_holds': True, 'counterexample': ''
TRIAL: {'metric_name': 'Spearman rank correlation', 'metric_value': 1.0, 'instances_tested': 1, 'n_max': 27, 'conjecture_holds': True, 'counterexample': ''}

```

```

TRIAL: {'metric_name': 'Spearman rank correlation', 'metric_value': 1.0, 'instances_tested': 1, 'n_max': 38}
TRIAL: {'metric_name': 'Spearman rank correlation', 'metric_value': 1.0, 'instances_tested': 1, 'n_max': 38}
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TRIAL: {'metric_name': 'Spearman rank correlation', 'metric_value': 1.0, 'instances_tested': 1, 'n_max': 38}
TRIAL: {'metric_name': 'Spearman rank correlation', 'metric_value': 1.0, 'instances_tested': 1, 'n_max': 38}
RESULT: SUPPORTED mean=1.0 std=0.0 support_fraction=1.0

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test code only runs trials with a maximum of $n_{\max} = 38$, which is too small to validate the conjecture for larger instances. The metric does not scale trivially with n , but the limited testing range raises concerns about generalizability.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code only runs trials with a maximum of $n_{\max} = 38$, which is below the pre-registered criterion's threshold and raises concerns about generalizability. | next: Increase the range of tested instances to meet the pre-registered support condition and validate the conjecture for larger instances.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	18248
2	propose	ollama_remote	glm4:latest	0	0	28826
3	preregistration	ollama_remote	glm4:latest	0	0	9846
4	novelty	ollama_remote	glm4:latest	0	0	10960
5	novelty	ollama_remote	glm4:latest	0	0	9566
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18047
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	60501
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	61810
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	21837
10	critic	ollama_remote	glm4:latest	0	0	18533
11	judge	ollama_remote	glm4:latest	0	0	19101

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 277276 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in [research/audit/687e17de88a9.jsonl](#))

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/687e17de88a9.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/687e17de88a9.tar.gz` (if generated)